UNIVERSITY OF TWENTE.

Space & Time

In-depth Topic Assignment

Clustering of Point Data

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1 Introduction

The COVID-19 pandemic spread across countries and regions through complex spatial and temporal processes. In the Netherlands, the epidemic unfolded unevenly across municipalities, with some areas experiencing early outbreaks while others saw delayed increases in infection counts, a pattern observed in many countries during the COVID-19 pandemic (Kraemer et al., 2020). Understanding how such epidemics propagate across space is important for evaluating public health responses and for improving epidemiological modelling. One concept frequently used to describe the spatial spread of innovations and diseases is hierarchical diffusion, where processes initially spread through highly connected urban centres before gradually reaching smaller or more peripheral regions (Hägerstrand, 1967; Rogers, 2003).

Hierarchical diffusion is particularly relevant for infectious diseases because human mobility networks and population density strongly influence transmission dynamics (Balcan et al., 2009; Brockmann & Helbing, 2013). Large cities often act as hubs of connectivity, with greater international travel, higher population densities, and more intensive social interaction.

As a result, epidemics may first intensify in these urban areas before spreading outward to smaller municipalities. Detecting such spatial patterns requires analytical approaches capable of comparing epidemic trajectories across many locations simultaneously. The Netherlands provides a useful case study for examining such processes because its dense urban network and high levels of regional mobility create strong spatial connectivity between municipalities.

Time-series analysis provides a natural framework for studying epidemic dynamics. Municipality-level COVID-19 case counts form temporal sequences that reflect the progression of infection waves over time. However, comparing hundreds of time series directly is challenging because epidemic curves may differ in magnitude, timing, and noise levels. Time-series clustering methods provide a useful solution by grouping municipalities that exhibit similar temporal epidemic patterns (Liao, 2005). By identifying clusters of similar epidemic trajectories, it becomes possible to detect typical temporal profiles, such as early-peaking, intermediate, or delayed epidemic patterns. When combined with spatial mapping of cluster membership, these patterns can reveal whether epidemics spread through hierarchical or geographically structured diffusion processes.

While many studies analyse national epidemic dynamics, fewer studies examine how municipality-level epidemic trajectories cluster temporally and spatially. Understanding these local patterns is important for identifying how epidemics propagate through regional mobility and settlement networks.

In this project, clustering techniques are applied to municipality-level COVID-19 time series from two different sources. The first dataset consists of observed case counts reported by the Dutch National Institute for Public Health and the Environment (RIVM). These data provide a record of how the epidemic evolved across Dutch municipalities over multiple waves of infection. The second dataset is generated by an agent-based model (ABM) of COVID-19 transmission. Agent-based models simulate disease spread by representing individuals and their interactions, allowing researchers to explore how behavioural patterns and contact networks influence epidemic outcomes (Epstein, 2009). The ABM produces simulated infection counts at the municipality level, which can be analysed in a manner similar to the real data.

Comparing the observed RIVM data with the simulated ABM output provides an opportunity to evaluate how well the model reproduces real epidemic dynamics. If the model realistically captures transmission processes and spatial interactions, the temporal clustering patterns observed in the simulation should resemble those found in the real data. Conversely, substantial differences between the two datasets may indicate limitations in the model structure or assumptions. Such comparisons are therefore useful for assessing the realism and explanatory power of epidemiological simulations.

To investigate these issues, the analysis follows a reproducible workflow that includes data restructuring, preprocessing, time-series segmentation, and clustering of municipality-level epidemic trajectories. The clustering results are then interpreted both temporally and spatially in order to identify possible diffusion patterns. Finally, the observed and

simulated clustering structures are compared to evaluate the extent to which the model reproduces the spatial and temporal organisation of the real epidemic.

This study is guided by three research questions:

1. What temporal patterns appear in municipality-level COVID-19 time series in the Netherlands?

2. Do RIVM observations and ABM simulations produce similar or different clustering structures?

3. Do the resulting clusters provide evidence of hierarchical diffusion in the spread of COVID-19 across municipalities?

By addressing these questions, the analysis aims to improve understanding of spatial epidemic dynamics and to assess the extent to which agent-based simulations can reproduce the temporal and geographical patterns observed in real epidemic data.

2 Data and Study Material

2.1 RIVM Data

The first dataset used in this study consists of municipality-level COVID-19 case counts reported by the Dutch National Institute for Public Health and the Environment (RIVM). These data provide an empirical record of the progression of the COVID-19 epidemic across municipalities in the Netherlands and allow construction of epidemic trajectories for individual locations.

The original RIVM data are provided in a long tabular format in which each row corresponds to a municipality–date combination. For time-series analysis and clustering, the data were restructured into a municipality $\times$ time matrix, where each row represents a municipality and each column corresponds to a time step in the epidemic. Each row therefore forms a time series describing the evolution of COVID-19 cases for that municipality.

The dataset covers the main period of the epidemic in the Netherlands, beginning in February 2020 and extending into 2023. After restructuring and aggregation, the time series contain approximately 150 weekly observations per municipality. Weekly aggregation was used to reduce reporting noise and ensure comparability with the temporal resolution used in the simulation outputs.

Following preprocessing, the dataset includes 342 municipalities with complete or near-complete time series used for clustering analysis of observed epidemic dynamics.

2.2 ABM Simulation Data

The second dataset consists of outputs generated by an agent-based model (ABM) simulating the spread of COVID-19 in the Netherlands. Agent-based models represent individuals as agents interacting within a simulated environment, allowing epidemic dynamics to emerge from behavioural interactions and contact patterns. Such models are commonly used to explore the effects of interventions, mobility patterns, and contact structures on disease transmission.

In the simulation used for this study, infection counts are recorded at the municipality level over time. Similar to the RIVM data, the ABM output was transformed into a municipality $\times$ time matrix, where each municipality is represented by a time series of simulated infection counts.

The ABM dataset contains approximately 396 municipalities, reflecting the spatial resolution used in the simulation environment. Simulation outputs provide infection counts at regular time intervals and were aggregated to weekly values to ensure comparability with the RIVM dataset. This step allows both datasets to be analysed using the same temporal resolution.

The resulting time series capture the modelled epidemic trajectory for each municipality and enable comparison between observed epidemic patterns and simulated dynamics.

2.3 Analytical Workflow

To analyse and compare the observed and simulated epidemic dynamics, a structured analytical workflow was developed. The workflow transforms raw data into a form suitable for time-series clustering and spatial interpretation.

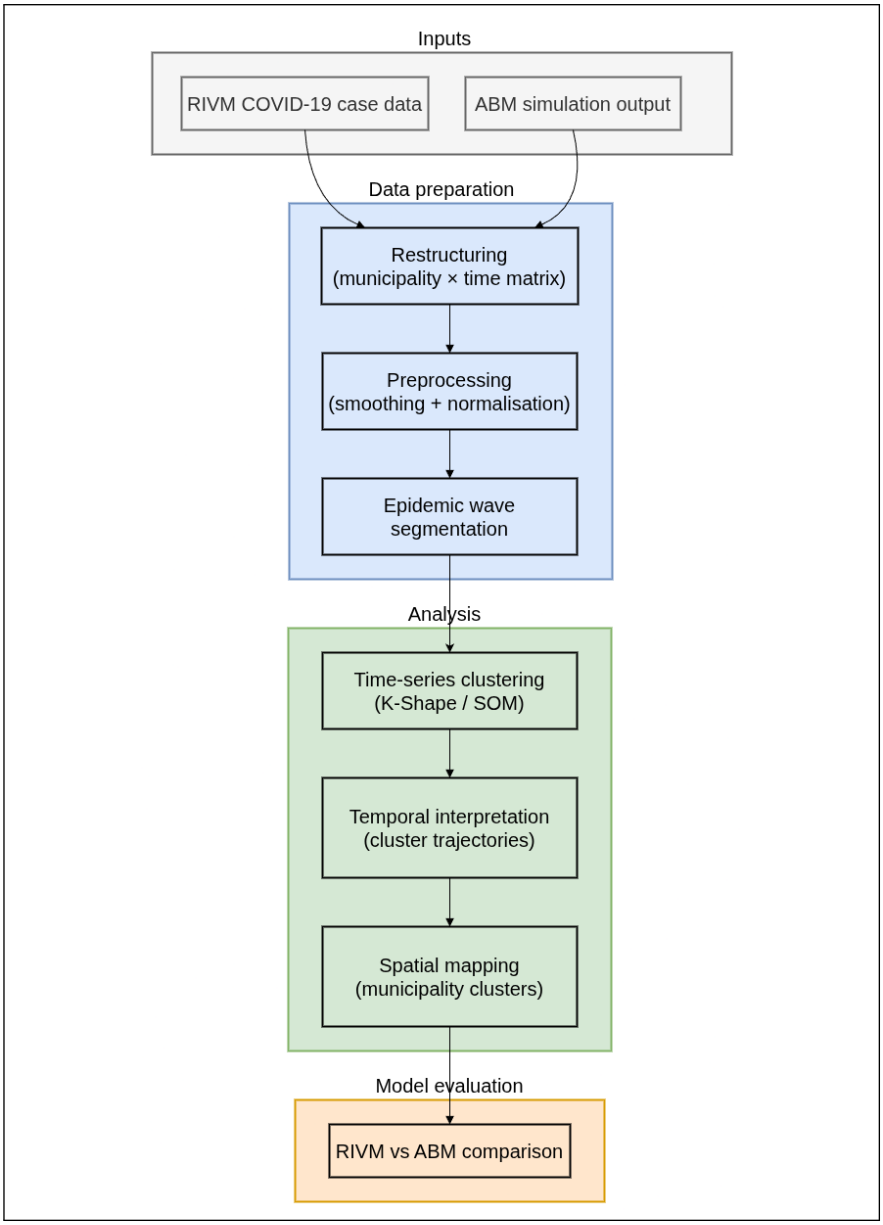

Figure 1: Analytical workflow

Figure 1 illustrates the main stages of the analysis pipeline. The workflow begins

with the two data sources, RIVM observations and ABM simulation outputs. These datasets are first restructured into municipality × time matrices. The time series are then preprocessed through smoothing and normalisation in order to reduce noise and emphasise temporal patterns.

Next, the epidemic timeline is segmented into major waves to support interpretation of different epidemic phases. Time-series clustering methods are then applied to group municipalities with similar epidemic trajectories. The resulting clusters are analysed both temporally and spatially, allowing the identification of diffusion patterns and regional epidemic structures. Finally, clustering results from the RIVM and ABM datasets are compared in order to evaluate how well the simulation reproduces the observed epidemic dynamics.

Table 1 summarises the key characteristics of the two datasets used in the analysis.

Table 1: Dataset overview

S.N.	Dataset	Municipalities	Time steps	Date range	Temporal resolution
1	RIVM	344	84	Mar 2020 – Oct 2021	Weekly
2	ABM	396	79	Simulation period	Weekly

This overview highlights the similarities in temporal structure between the datasets, which enables meaningful comparison of clustering results across observed and simulated epidemic time series.

3 Methodology

All data preprocessing, clustering procedures, and visualisations were implemented in Python and documented in the accompanying analysis notebook.

3.1 Data Architecture and Preprocessing

To enable comparison between observed and simulated epidemic dynamics, both datasets were organised into a consistent municipality × time matrix structure. In this format, each row represents a municipality and each column represents a time step, allowing infections to be analysed as time series for each spatial unit.

The original RIVM dataset was provided in long format, where each record contained a municipality identifier, date, and reported number of COVID-19 cases. For time-series analysis, the data were reshaped using a pivot operation so that municipalities formed the rows and weekly time steps formed the columns. This produced a matrix in which each row represents the epidemic trajectory of a municipality across the observation period.

The ABM simulation output contained infection counts generated by the agent-based model for multiple municipalities. Although its structure differed slightly from the observational dataset, it was reorganised into the same municipality × time matrix to ensure compatibility with the analytical pipeline used for the RIVM data.

After restructuring, preprocessing was applied to reduce noise and emphasise epidemic patterns. Weekly infection counts may fluctuate because of reporting delays or stochastic simulation effects. A rolling mean smoothing procedure was therefore applied to each municipality time series to reduce short-term variability while preserving overall epidemic trends.

Finally, the time series were normalised using z-score standardisation. This ensures clustering focuses on the temporal shape of epidemic trajectories rather than differences in magnitude between municipalities.

3.2 Segmentation

Epidemic processes often occur in multiple waves rather than as a single continuous outbreak. To account for this structure, the analysis incorporated temporal segmentation based on epidemic waves.

Wave detection was performed using the national epidemic curve, obtained by aggregating infection counts across all municipalities. The national curve provides a high-level representation of the overall epidemic progression and makes major peaks and troughs in infection levels clearly visible.

Using the smoothed national curve, major epidemic phases were identified by examining the timing of peaks and transitions between periods of rising and declining case numbers. These turning points were used to define analysis segments corresponding to different epidemic waves. Each segment therefore represents a period during which the epidemic followed a relatively coherent temporal pattern across the country.

Segmenting the epidemic into waves serves two purposes. First, it allows clustering algorithms to focus on specific epidemic phases, which may reveal patterns that are less visible when analysing the entire time series. Second, segmentation makes it possible to compare how municipalities behaved during different waves, highlighting variations in the timing or intensity of local outbreaks.

Both the full time series and wave-based segments were considered in the analysis to evaluate how segmentation affects clustering results and interpretability.

3.3 Clustering Methods

To identify groups of municipalities with similar epidemic trajectories, two time-series clustering approaches were considered: K-Shape clustering and Self-Organising Maps (SOM).

K-Shape clustering is specifically designed for time-series data and uses shape-based similarity measures to group time series according to the form of their temporal patterns. The method aligns time series by maximising the similarity of their shapes, making it particularly suitable for analysing epidemic curves where the timing and structure of peaks are more important than absolute magnitude. K-Shape therefore emphasises similarities in the progression of epidemic waves across municipalities.

In addition to K-Shape, Self-Organising Maps (SOM) were explored as an alternative clustering approach. SOM is an unsupervised neural network algorithm that projects high-dimensional data onto a lower-dimensional grid while preserving topological relationships between observations. In the context of time-series analysis, SOM can reveal clusters of municipalities with similar temporal patterns while also providing an interpretable visual representation of the clustering structure.

The two clustering methods rely on different similarity concepts. K-Shape focuses on shape-based similarity, typically derived from cross-correlation between time series, while SOM commonly uses Euclidean distance in the transformed feature space. Because Euclidean distance is sensitive to magnitude differences, the z-score normalisation step described earlier was important to ensure that clustering primarily reflected temporal patterns rather than absolute case counts.

By applying different clustering approaches and similarity measures, the analysis evaluates how methodological choices influence the identification of epidemic patterns.

3.4 Cluster Selection and Visualisation

An important step in clustering analysis is determining the appropriate number of clusters. Rather than fixing this number in advance, several candidate solutions were tested with cluster numbers ranging from three to six. This range balances the need to identify meaningful patterns while avoiding overly fragmented cluster structures.

To evaluate these candidate solutions, both quantitative and qualitative criteria were used. The silhouette score, a commonly used internal evaluation metric, was applied to measure the separation between clusters and the cohesion of observations within each cluster. Higher silhouette scores indicate better-defined clusters.

However, numerical metrics alone are not always sufficient for time-series clustering, particularly in epidemiological applications where interpretability is important. Cluster solutions were therefore also evaluated based on the interpretability of temporal profiles. Centroid curves were examined to determine whether clusters represented distinct epidemic trajectories, such as early, intermediate, or late epidemic peaks.

Spatial coherence was also considered. After clustering, municipalities were mapped to assess whether clusters displayed meaningful geographic patterns, such as concentrations in urban regions or peripheral areas.

By combining silhouette scores, centroid interpretability, and spatial coherence, a final clustering configuration was selected for both the RIVM and ABM datasets.

4 Results

4.1 Preprocessing and Segmentation Results

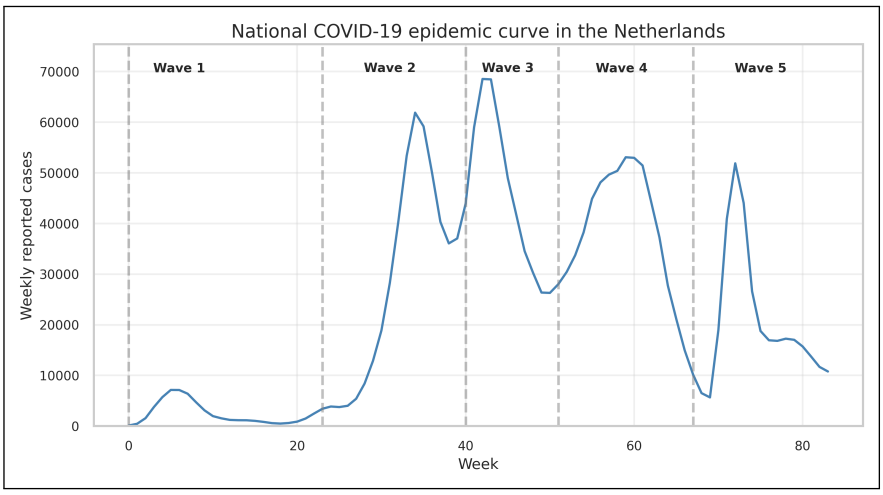

Figure 2: National epidemic curve with wave boundaries

Figure 2 shows the national COVID-19 epidemic curve derived by aggregating municipality-level case counts across the Netherlands. The curve illustrates several distinct epidemic waves between early 2020 and late 2021. Based on the national trajectory, wave boundaries were defined to segment the epidemic into major phases of transmission.

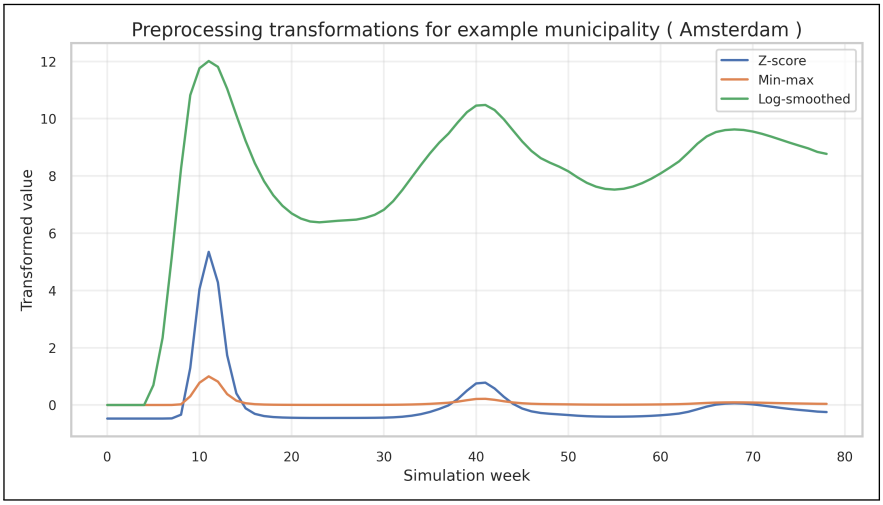

Figure 3: Preprocessing transformations for example municipality (Amsterdam)

The effect of these preprocessing transformations is illustrated in Figure 3, which shows an example municipality (Amsterdam) under different representations, including z-score normalisation, min–max scaling, and a log-smoothed transformation. These transformations illustrate how preprocessing alters the scale and shape of municipality epidemic trajectories.

4.2 Clustering Results for RIVM

Clustering of municipality-level COVID-19 time series from the RIVM dataset revealed several distinct epidemic trajectories across the Netherlands. These clusters represent groups of municipalities that experienced similar temporal patterns in reported infections. The clustering analysis highlights differences in the timing and intensity of epidemic waves across regions.

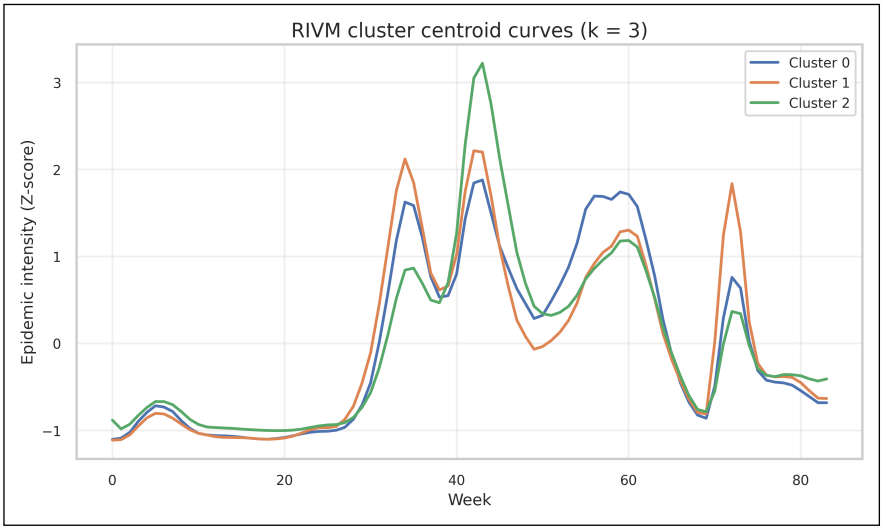

Figure 4: RIVM cluster centroid curves

Figure 4 presents the centroid curves for the clusters identified from the RIVM time series. Each centroid represents the average temporal trajectory of municipalities assigned to that cluster after preprocessing and normalisation using a z-score transformation. The centroid curves show clear differences in the timing and magnitude of epidemic peaks between clusters. While all clusters exhibit the major national waves, their peak timing and relative intensity vary. One cluster displays stronger peaks during the mid-phase waves of the epidemic, whereas another shows relatively delayed or more moderate peaks. These differences indicate temporal heterogeneity in how municipalities experienced successive epidemic waves.

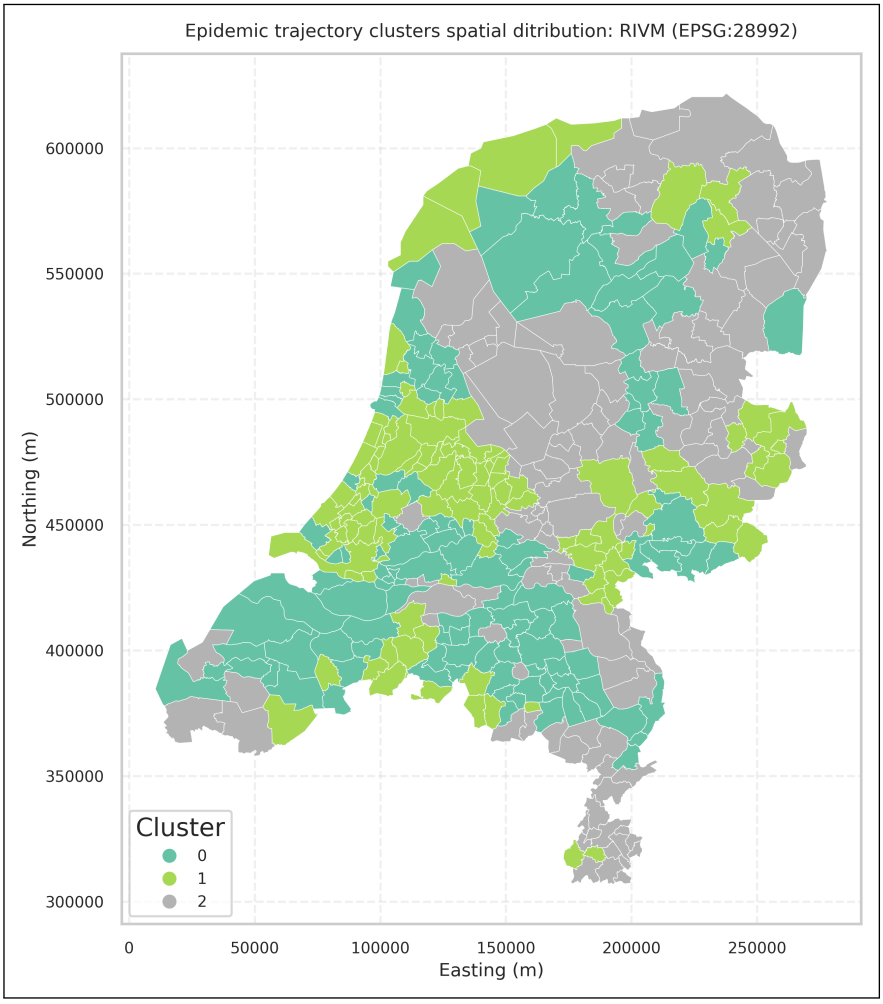

Figure 5: RIVM spatial cluster map

The spatial distribution of the clusters is shown in Figure 5. Municipalities belonging to different clusters are distributed across the Netherlands, with some regional groupings visible but no strictly uniform geographic separation. Certain clusters appear more common in parts of the western and central Netherlands, while others are more frequently observed in northern and southern municipalities. However, the spatial distribution remains mixed, with neighbouring municipalities sometimes belonging to different clusters.

4.3 Clustering Results for ABM

Clustering of the municipality-level time series generated by the agent-based model (ABM) revealed several epidemic trajectory patterns across the simulated municipalities. The clustering groups municipalities that exhibit similar temporal dynamics in infection counts throughout the simulation period.

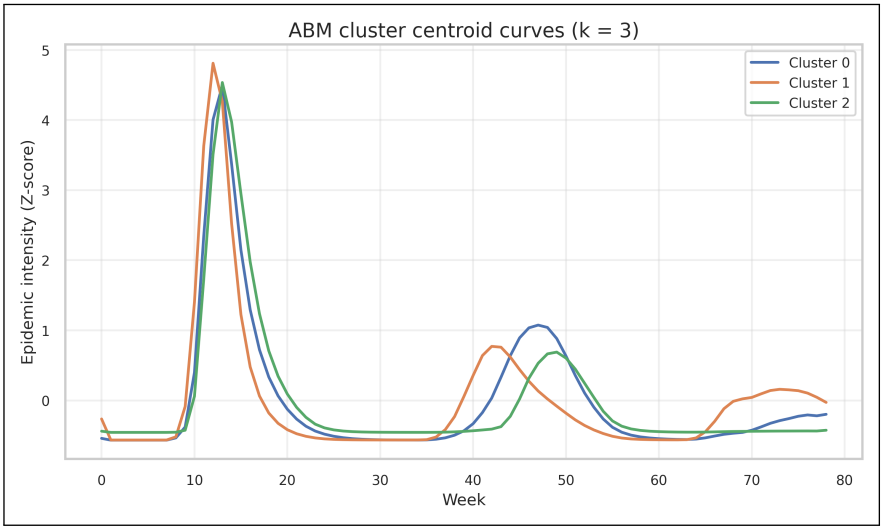

Figure 6: ABM cluster centroid curves

Figure 6 presents the centroid curves for the clusters identified from the ABM time series. Each centroid represents the average temporal trajectory of municipalities assigned to a cluster after preprocessing and normalisation using a z-score transformation. The centroid curves indicate that the simulated epidemic waves occur at broadly similar times across clusters. The initial epidemic peak appears highly synchronised across all clusters, occurring within a narrow time window early in the simulation. Later waves also occur within comparable periods, although modest differences in peak height and slight shifts in timing are visible between clusters.

The simulated epidemic trajectories follow a relatively consistent temporal evolution across municipalities. Differences between clusters primarily reflect variation in epidemic intensity rather than large differences in the timing of infection waves. As a result, the temporal variation between clusters is relatively limited.

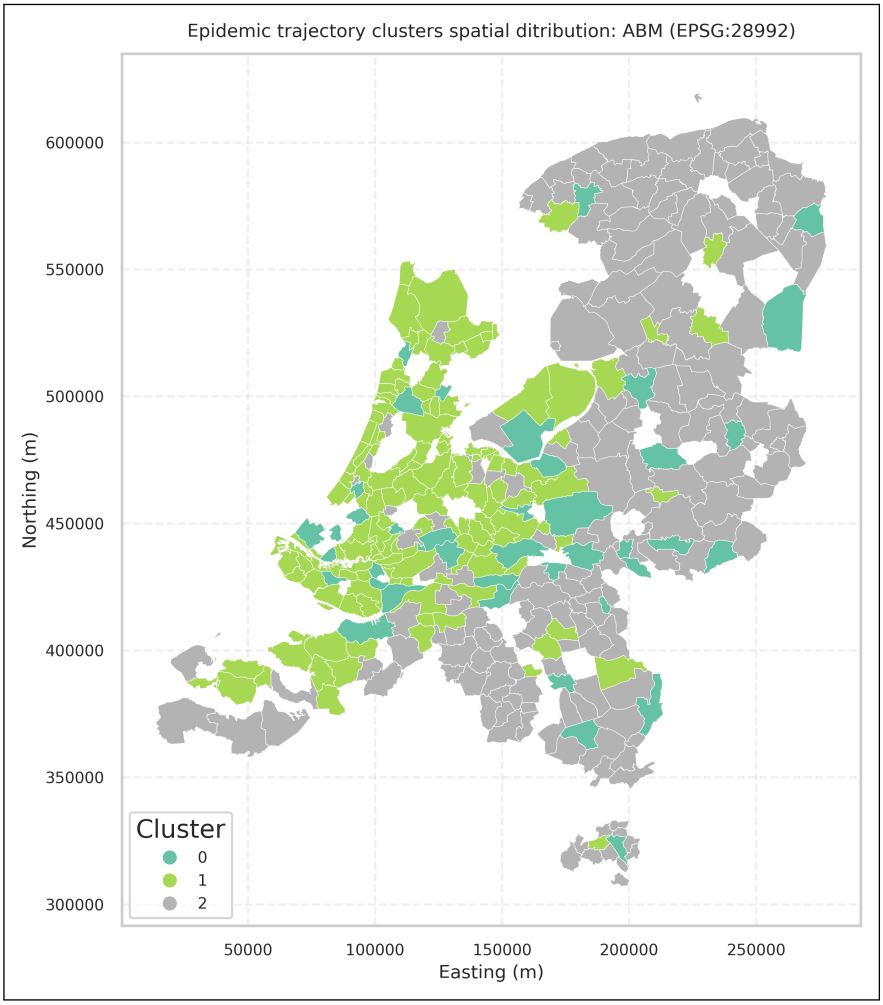

Figure 7: ABM spatial cluster map

The spatial distribution of the ABM clusters is shown in Figure 7. The spatial cluster map indicates that municipalities belonging to different clusters are distributed across the Netherlands without a clear or continuous geographic ordering. Although some clusters appear more common in particular regions, municipalities from different clusters are interspersed throughout the country.

4.4 Comparative Analysis

The clustering results obtained from the RIVM and ABM datasets reveal important differences in the temporal and spatial structure of epidemic trajectories across municipalities. While both datasets produce identifiable trajectory clusters, the patterns observed in the empirical and simulated data differ in several key aspects.

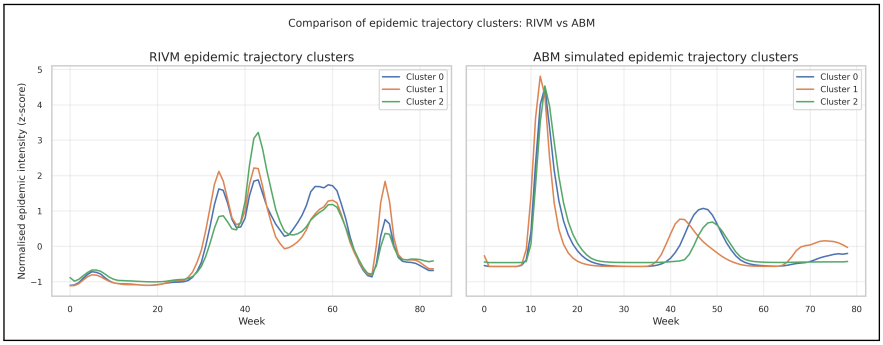

Figure 8: Comparison of epidemic trajectory clusters: RIVM and ABM

Figure 8 presents a comparison of the centroid curves derived from the RIVM and ABM clustering analyses. The centroid curves for the RIVM data exhibit clear variation in the timing and magnitude of epidemic peaks between clusters. Although the major national waves are present across all clusters, the timing of peak infection intensity differs across groups of municipalities. Some clusters show earlier increases in case numbers during particular waves, while others display delayed peaks or more moderate trajectories. This staggered structure indicates substantial temporal heterogeneity in the progression of the epidemic across municipalities.

In contrast, the centroid curves derived from the ABM simulation show a more synchronised pattern. The major epidemic peaks appear within similar time windows across clusters, with only moderate variation in peak intensity. While some clusters exhibit slightly higher or lower peaks, the overall timing of epidemic waves remains largely aligned across municipalities. This suggests that the simulated epidemic dynamics evolve in a more uniform manner than the observed trajectories in the empirical data.

Spatial patterns also differ between the two datasets. The RIVM clustering results show partial regional structure, with certain clusters appearing more frequently in particular parts of the country. By contrast, the ABM spatial clusters display a more mixed distribution across the Netherlands, with municipalities from different clusters appearing interspersed throughout the map. This indicates weaker geographic organisation in the simulated epidemic trajectories.

Table 2 summarises the key differences observed between the clustering outcomes for the two datasets.

Table 2: Cluster comparison summary

S.N.	Aspect	RIVM	ABM	Interpretation
1	Temporal patterns	Distinct early–late cluster trajectories	Similar peak timing across clusters	Real epidemic shows stronger temporal heterogeneity
2	Spatial concentration	Some regional concentration of clusters	Mixed spatial distribution	Weaker geographic ordering in ABM
3	Hierarchical diffusion	Patterns consistent with hierarchical diffusion	Limited evidence of diffusion structure	Model spreads infection more uniformly

4.5 Sensitivity Analysis

To evaluate the robustness of the clustering design, a sensitivity analysis was conducted using alternative preprocessing strategies applied to the epidemic time series. The objective was to assess how different temporal representations influence the quality and interpretability of trajectory clusters.

Table 3: Sensitivity analysis results

S.N.	Setting	Result quality	Comment
1	Raw full-series clustering	Weak	Noise in raw case counts reduces cluster separation
2	Smoothed full-series clustering	Moderate	Smoothing highlights epidemic wave structure
3	Wave-segmented clustering	Strong	Wave segmentation improves phase-specific cluster separation

Table 3 summarises the results of this analysis. Three configurations were tested: clustering using the raw full time series, clustering after temporal smoothing, and clustering based on segmented epidemic waves.

Clustering performed on the raw full series produced weak results. The trajectories contained substantial short-term fluctuations and reporting noise, which reduced the clarity of cluster separation and made centroid patterns difficult to interpret.

Applying temporal smoothing improved the results by reducing high-frequency variability and revealing the main epidemic trends. The resulting clusters showed clearer trajectory shapes and more stable centroid curves.

The strongest clustering performance was obtained when the time series were segmented into epidemic waves before clustering. This approach isolates major phases of the epidemic and reduces noise between waves, producing the clearest separation between trajectory groups.

Overall, the analysis indicates that wave segmentation provides the most robust representation of epidemic dynamics for clustering.

5 Discussion

This section interprets the clustering results in relation to spatial diffusion theory, epidemic modelling, and methodological considerations.

5.1 Evidence for Hierarchical Diffusion

The clustering analysis of the RIVM dataset reveals differences in the timing and structure of epidemic trajectories across municipalities. The centroid curves show that epidemic peaks did not occur simultaneously across the country. Instead, clusters exhibit staggered peaks, with some municipalities experiencing earlier increases in infection counts while others show delayed trajectories. This pattern indicates temporal heterogeneity in how the epidemic progressed across municipalities.

Such staggered dynamics are consistent with hierarchical diffusion. In this framework, infections initially intensify in highly connected urban centres that function as transmission hubs within mobility networks. From these centres, the epidemic spreads outward through human movement and interaction, gradually reaching smaller settlements.

The spatial distribution of clusters supports this interpretation. Earlier-peaking clusters appear more frequently in the highly urbanised Randstad region, where mobility connectivity and population density are high, while later-peaking clusters occur more

often in peripheral municipalities. These results suggest epidemic propagation followed patterns consistent with hierarchical diffusion through urban connectivity networks.

5.2 Real and Simulated Epidemic Structure

Comparison of the clustering results reveals important differences between the empirical epidemic patterns observed in the RIVM data and those produced by the agent-based model. In the RIVM dataset, the centroid curves display clear variation in the timing of epidemic peaks between clusters. Different groups of municipalities experienced epidemic waves at different moments, producing a staggered temporal pattern across the country and indicating substantial temporal heterogeneity in epidemic progression.

In contrast, the centroid curves derived from the ABM simulation show a more synchronised structure. Epidemic peaks occur within similar time windows across clusters, with only modest variation in peak magnitude. Spatial patterns are also weaker, with clusters distributed more evenly across municipalities rather than forming distinct regional groupings.

One possible explanation is that the simulation assumes relatively homogeneous mobility or contact structures between municipalities. If interaction rates or mobility connections are represented too uniformly, infections can propagate through the network more evenly, producing synchronised epidemic waves. Simplified assumptions about contact mixing or demographic variation may further reduce local differences in epidemic timing.

These results suggest that while the ABM captures general epidemic waves, it may underrepresent regional diffusion dynamics observed in real epidemic data.

5.3 Methodological Implications

Methodological choices influenced the interpretability of the clustering results. First, smoothing of municipality-level time series reduced short-term fluctuations caused by reporting delays and local variability. Weekly COVID-19 case counts often contain noise that obscures epidemic trends, and applying a rolling mean filter made the broader structure of epidemic waves clearer.

Normalisation using a z-score transformation was also essential. Municipalities differ greatly in population size and case counts, which can dominate clustering results. Standardisation ensured that clustering focused on the timing and shape of epidemic trajectories rather than magnitude.

Temporal segmentation of the epidemic into major waves further improved clustering clarity. Analysing epidemic phases allowed the clustering algorithm to detect differences in municipal behaviour during specific waves.

Finally, the K-Shape algorithm proved suitable because it groups time series based on similarity in temporal shape, making it appropriate for analysing epidemic curves.

5.4 Limitations

Several limitations should be considered when interpreting these results. First, early stages of the COVID-19 pandemic in the Netherlands were affected by limited testing capacity. Consequently, reported RIVM case counts likely underestimate true infection levels during initial waves, which may influence the observed trajectories.

Second, vaccination dynamics were not incorporated in the analysis. Increasing vaccination coverage during later phases of the pandemic may have influenced infection patterns and the timing of later waves.

Third, the analysis was conducted at the municipality level. Aggregating data at this scale may conceal finer spatial transmission patterns within municipalities.

Finally, clustering outcomes depend on the chosen number of clusters. Different values of k could produce alternative grouping structures and different interpretations of epidemic trajectories.

5.5 Recommendations

Future analyses of epidemic diffusion could be improved in several ways. Using population-adjusted incidence rates instead of raw case counts would improve comparability between municipalities, while incorporating vaccination dynamics would better capture changing population susceptibility.

Methodologically, similarity measures such as Dynamic Time Warping (DTW) and network-based clustering could better represent temporal shifts and spatial connectivity.

From a modelling perspective, incorporating mobility networks and greater spatial heterogeneity in the ABM may produce more realistic regional diffusion patterns.

6 Conclusion

This study analysed municipality-level COVID-19 time series in the Netherlands using time-series clustering to investigate spatial diffusion patterns and compare empirical epidemic data with agent-based model (ABM) simulations.

The clustering results from the RIVM dataset revealed clear temporal heterogeneity in epidemic trajectories across municipalities. Distinct clusters showed differences in the timing and intensity of epidemic waves, with some municipalities experiencing earlier peaks than others. The spatial distribution of these clusters suggests that earlier trajectories occurred more often in highly connected urban regions, while later trajectories appeared more frequently in peripheral municipalities. These patterns are consistent with hierarchical diffusion, in which infections spread progressively from major urban centres to smaller settlements.

The ABM simulation reproduced some epidemic dynamics but generated more synchronised trajectories across municipalities than those observed in the empirical data. Cluster centroid curves indicate that epidemic peaks occurred within similar time windows across simulated clusters, reflecting weaker temporal variation than in the empirical data.

Methodologically, the results show that preprocessing strongly influences clustering outcomes. Wave-based segmentation combined with smoothing and z-score normalisation produced the most interpretable trajectory clusters by highlighting epidemic wave structure while reducing noise.

These findings highlight the importance of incorporating realistic mobility structures and spatial heterogeneity when using agent-based models to simulate epidemic diffusion.

7 Use of AI

ChatGPT was used for debugging and commenting the Jupyter Notebook code, and for removing spelling and grammar errors in the report.

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